

Alexander Kaneko

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EDUCATION

Brown University

Providence, RI

Bachelor of Science in Applied Mathematics and Computer Science

2028

- Relevant Coursework: Deep Learning, Theoretical Machine Learning, Database Systems, Computer Systems, Computational Molecular Biology, Linear Algebra, Probability and Statistics, Abstract Algebra

EXPERIENCE

Researcher

June 2026 – Sept 2026

Statistical Artificial Intelligence Lab – WPI

- Developed a diffusion-model-based system to reconstruct ECG waveforms from PPG signals, using Bayesian posterior sampling to quantify uncertainty and attribute it to distinct physiological and model sources; contributed to the theory and experiments for an ICASSP submission.
- Reimplemented and benchmarked two state-of-the-art baselines in PyTorch from publications lacking released training code, diagnosing a fundamental failure mode in the published method and delivering a validated comparison that matched reported accuracy on clinical data.

Machine Learning Intern

May 2026 – Aug 2026

UMass Chan Medical School – Colubri Lab

- Benchmarked 10 model configurations across 5 families (PyTorch GRUs, logistic regression, random forest, extra trees, gradient boosting) for next-day illness prediction on 4,679 traveler-days of clinical time-series data; top models reached 0.52 PR-AUC, a 30 % lift over the persistence baseline ($p < 0.0001$).
- Designed a stratified k-fold validation harness with paired significance testing and positive controls, surfacing a trip-level data leak that had inflated ROC-AUC by 0.05, a full-batch training defect costing 0.025 PR-AUC ($p=0.05$, 8/10 folds), and 0.015 PR-AUC of hyperparameter selection bias.

Teaching Assistant

Jan 2026 – May 2026

Brown University

- Selected as a TA for CSCI 0410: Fundamentals of AI
- Assisted with course content, led sections, and held office hours

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript/TypeScript, PostgreSQL, LaTeX, HTML/CSS

Frameworks: React, Node.js, Next.js, JUnit, LangChain

Developer Tools: Git, GitHub, XCode, VS Code, Vercel, IntelliJ, Figma, Supabase

Libraries: numpy, pandas, Tailwind, NextAuth.js, Cosmic CMS

PROJECTS

Blackjack Decision Modeling | *Python, Jupyter*

2025

- Built a configurable Blackjack simulation engine supporting realistic casino rules (splitting, doubling, variable bets, multi-hand play) for large-scale strategy evaluation
- Implemented value-based policy evaluation to compute expected returns and benchmarked performance as state complexity increased, identifying practical limits of exact planning
- Developed and tested reinforcement learning-based agents as scalable alternatives, comparing full-state policies against compact approximations such as card-count signals

GuitarHero | *Next.js, LangChain, TypeScript, PostgreSQL*

2025

- Built a modern web app for guitarists with real-time chord diagrams, synchronized lyrics, and Spotify playback integration using the Spotify Web Playback SDK and NextAuth.js
- Integrated an AI-powered chatbot using LangChain and the OpenAI API to provide personalized chord guidance, song-specific practice tips, and on-demand theory explanations, improving user engagement and learning efficiency by over 30%
- Engineered interactive chord learning tools and community features, including user-contributed chord arrangements, admin verification, and genre-based song exploration, backed by Supabase (PostgreSQL)